Clear-HR: Holiday & Absence Management — Software Design Document

Overview

Holiday management for ClearHR allows organisations to configure and manage employee time off. The system is designed to be flexible: organisations choose how holidays are measured, how they're allocated, and whether employees request approval, book directly or an Admin can book directly. Bank Holidays can be added to the Personal Allowance or booked automatically.

Glossary

Term	Definition
Absence Reason	The reason for taking an absence. E.g. ‘Running a temperature’
Absence Type	A grouping for Absence Reasons e.g. ‘Sick Unpaid’
Allowance	Synonym for entitlement — used interchangeably.
Accrual	Entitlement that accumulates gradually over time (e.g. monthly) rather than being available upfront.
Borrow-Ahead	Amount of holiday that can be taken from next period’s allowance
Carry-over	Unused holiday from one period that transfers into the next. May be capped or time-limited.
Entitlement	The total amount of holiday an employee is allocated for a given period (e.g. 25 days per year).
Flexible	A Holiday type where holidays are earned depending on an employee’s previous work pattern

Holiday year	The period over which entitlement is measured. Typically Jan–Dec or Apr–Mar, configurable per org.
Request	An employee asks for time off; an approver must accept or reject it before it's confirmed.
Booking	Certain employees can book time off directly without approval (self-service).
Approver	A member (typically admin or team lead) with can_approve_holidays permission who can accept/reject requests.
TOIL	Time Off In Lieu — compensatory time off earned by working extra hours.
Bank holiday	A public/national holiday. Added to an employee's allowance or booked automatically.
Black List	Dates when Holidays cannot be taken by default
White List	Dates when Holidays must be taken by default
Working pattern	Which days of the week an employee works (e.g. Mon–Fri, or Mon/Wed/Fri). Affects how "days" are counted.
Half day	A partial-day absence (AM or PM).
Overlap	Two approved absences for the same employee on the same date — typically blocked.
Profile	An Org can create an unlimited number of Holiday/Absence Profiles. Each Employee will be allocated to a Profile. Each Profile will determine such parameters as whether the Holidays are calculated in Days or Hours, Fixed or Flexible and Carry Over Rules.
Tracked	A property of an Absence Type that determines if absence reasons of this type have an Absence Profile with allowance, accrual and carry forward.

Holiday Profiles

These are org-level settings that shape how an employee's holiday rules are enforced. Different Holiday Profiles can be applied for the same employee in different Periods. Each dimension is independent. Employees can move from one profile to another. For the Minimum Viable Product this change will be handled manually. Subsequently, tools will be developed to make the change automatic (with oversight).

Measurement: Days vs Hours

Setting	Days mode	Hours mode
Entitlement expressed as	25 days	200 hours
Booking granularity	Full day / half day (AM/PM)	Hours and minutes (e.g. 2h 30m)
Deduction from balance	1 day or 0.5 day	Actual hours booked
Working pattern relevance	Determines which days count	Determines daily hour capacity
Display in calendar	Day blocks	Time ranges

Allocation: Fixed vs Flexible

Setting	Fixed	Flexible
Entitlement available	Full amount from day one of the holiday year or accrued over time	Employees are allowed e.g. 5.6 weeks holiday a year. Hours per week depends on average hours worked in the previous 12 months
New starter mid-year	Pro-rated based on start date or Holiday Year or starts at employee start date.	Accrues naturally from start date
Visibility to employee	"You have 25 days this year" or "You have accrued 12.5 of 25 days so far"	"You have accrued 47 hours (9 days) which are available to use"

The Holiday Profile will determine whether to allow booking against future accrual (i.e. borrow ahead), or limit to accrued balance only. Default is to Limit.

Carry-over

Setting	Options
Allowed	Yes / No
Cap	Unlimited / Fixed max (e.g. 5 days)
Expiry	None / Use-by period (e.g. "carried days expire 3 months after period end)
Applies to	Holiday Profile

Borrow ahead

Setting	Options
Allowed	Yes / No
Cap	Fixed max (e.g. 5 days) default 0
Applies to	Holiday Profile

Request Profiles

Contain Rules specifying

Notice needed for absences (e.g. 10 days for up to 5 days absence, 2 days notice for < 5 days absence)

Minimum Team cover (e.g. can’t request absence if > 3 employees in the same team are absent or expressed as a percentage of employees absent)

Workflow: Request vs Book

Setting	Request (approval required)	Book (self-service)
Employee action	Submits a request	Confirms a booking directly
Approval step	Approver accepts or rejects	None — booking is immediate
Status flow	Pending > Approved / Rejected	Confirmed
Notifications	Approver notified on submit; employee notified on decision	Approver/team notified after booking
Cancellation	Employee can cancel pending; approved may need re-approval	Employee can cancel freely (within policy)

Features

Org Settings & Configuration

- [ ] Default Holiday year start date. Fixed day of month (default: 1st January) or Employee Start Date.
- [ ] bank_holiday_handling = 'additional' or 'deducted'

Holiday Profiles

- [ ] Measurement mode: days or hours (default Days)
- [ ] Allocation mode: fixed or Zero hours (default Fixed)
- [ ] Default workflow: request or book
- [ ] Max Carry-over rules (allowed, cap, expiry)
- [ ] Min Carry-over rules (allowed, cap)
- [ ] Adjust: Added to Allowance
- [ ] Borrow-ahead amount

AbsenceTypes

- [ ] Default absence types created on org setup (Annual Leave, Sick, Compassionate)
- [ ] Custom leave types (org can add/edit/delete)

- [ ] Per-type settings: paid/unpaid, tracked, requires approval, colour

Absence Reasons

- [ ] Belong to an Absence Type
- [ ] The entity that is requested or booked

Entitlements

- [ ] Applies per employee per absence type
- [ ] Pro-rating for mid-year starters/leavers
- [ ] Accrual calculation (monthly/weekly)
- [ ] Carry-over calculation at year-end
- [ ] Entitlement adjustments (manual add/subtract with reason)

Working Patterns

- [ ] Org-wide default working pattern (e.g. Mon–Fri)
- [ ] Per-employee working pattern override
- [ ] Part-time patterns (e.g. Mon/Wed/Fri)
- [ ] Hours per day (for hours-mode orgs)
- [ ] Working pattern history (changed mid-year — affects Flexible allowance)

Requesting / Booking

- [ ] Employee submits holiday request (date range, leave type, optional note)
- [ ] Half-day selection (AM/PM) in days mode
- [ ] Hours/time selection in hours mode
- [ ] Request Profile check before submission (warn or block if insufficient)
- [ ] Overlap detection (block or warn if employee already has leave on those dates)
- [ ] Self-service booking (when workflow = book)

Approver Profile

- [ ] Approval required
- [ ] Approval required: none, 1 or 2 approvers who are named admins
- [ ] Approval delegation (cover for absent approver)
- [ ] Approver sees pending requests (filtered to their scope — team or all)
- [ ] Approve / Reject with optional note
- [ ] Bulk approve
- [ ] Escalation (auto-remind or escalate if not actioned within N days)

Cancellation

- [ ] Employee cancels pending request
- [ ] Employee cancels approved holiday (will need re-approval)
- [ ] Admin/owner cancels on behalf of employee
- [ ] Cancelled days returned to balance

Calendar & Visibility

- [ ] Employee sees their own holiday calendar
- [ ] Team calendar (who's off when — based on permissions)
- [ ] Org-wide calendar (owners/admins)
- [ ] Bank holidays shown on calendar
- [ ] Colour-coded by leave type

Balances & Reporting

- [ ] Employee dashboard: entitlement, used, remaining, pending
- [ ] Admin dashboard: team balances at a glance
- [ ] Pending approvals
- [ ] Holiday report (filterable by team, date range, leave type)
- [ ] Export to CSV
- [ ] Integration with existing ClearHR reports system

Notifications

- [ ] Employee notified: request approved/rejected, upcoming holiday reminder
- [ ] Approver notified: new request pending
- [ ] Push notifications (mobile app)
- [ ] Email notifications

Mobile App

- [ ] View own balances and upcoming holidays
- [ ] Submit holiday request
- [ ] Cancel request/booking
- [ ] Push notification support
- [ ] view own calendar

Database Schema

To be populated as tables are created. Below is an initial sketch.

Tables (planned)

Absence_types — Configurable absence categories per org.

Column	Type	Notes
id	uuid	PK
organisation_id	uuid	FK to organisations
name	text	e.g. "Annual Leave"
requires_tracking	boolean	Has an allowance
is_paid	boolean	
deducts_from_entitlement	boolean	Sick leave may not deduct
requires_approval	boolean	Per-type workflow override
is_default	boolean	System-created, non-deletable
timestamps		

absence_reasons — Configurable absence reasons per org.

Column	Type	Notes
id	uuid	PK
organisation_id	uuid	FK to organisations

leave_type_id	uuid	FK to leave type
name	text	e.g. "PM Annual Leave"
colour	text	Hex colour for calendar
is_default	boolean	System-created, non-deletable
timestamps		

Absence_profiles - Configurable per Org

Column	Type	Notes
id	uuid	PK
organisation_id	uuid	FK
name	text	Name of this profile
absence_type_id	uuid	FK to absence_types
type	text	fixed/fixed accrued/flexible1/flexible2
carry_over_max	numeric	Max days/hours which can be carried over
carry_over_max_period	numeric	Number of days within which to take carried over holiday
carry_over_min	numeric	Min days/hours which can be carried over
borrow_ahead_max	numeric	Max days/hours which can be taken from next period
borrow_ahead_max_period	numeric	Number of days within which time can be borrowed from next period
Timestamps		

Holiday_year-record - Per-employee, per-leave-type, per-year allocation.

Column	Type	Notes
id	uuid	PK
organisation_id	uuid	FK
member_id	uuid	FK to members
absence_type_id	uuid	FK to absence_types
year_start	date	Start of the holiday year this covers (inclusive)
year_end	date	End of the holiday year this covers (inclusive)
base_amount	numeric	Org default or employee override (days or hours)
adjustment	numeric	Manual add/subtract (default 0)
carried_over	numeric	From previous year
borrow_forward	numeric	Number of hours/days pulled from next period
pro_rata_amount	numeric	Calculated effective entitlement after pro-rating
Timestamps		

work_profiles — working schedules.

Column	Type	Notes
id	uuid	PK
organisation_id	uuid	FK

name	text	E.g. ‘Weekly 37 hour’
member_id	uuid	FK to members (null = org default)
effective_from	date	When this pattern starts
hours_per_day	Numeric x7	For hours-mode orgs. 0 or null = not working
Timestamps		

employee_work_profiles — employee working schedules.

Column	Type	Notes
id	uuid	PK
work_profile__id	uuid	FK
member_id	uuid	FK to members
effective_from	date	When this pattern starts
Timestamps		

holiday_bookings — Individual absence records.

Column	Type	Notes
id	uuid	PK
organisation_id	uuid	FK
member_id	uuid	FK to members
leave_reason_id	uuid	FK to leave_reasons
start_date	date	

end_date	date	
start_half	text	null, 'am', 'pm' (days mode)
end_half	text	null, 'am', 'pm' (days mode)
hours_deducted	numeric	Total hours (hours mode)
days_deducted	numeric	Calculated days/hours deducted from balance
approver1_id	uuid	FK to members (who approved/rejected)
approver_note	text	
timestamps		

absence_comments - comments and status changes

Column	Type	Notes
id	uuid	PK
holiday_booking_id	uuid	FK to holiday_booking
member_id	uuid	Member making this comment
note	text	e.g. "Holiday in Spain"
holiday_status	text	'pending', 'approved', 'rejected', 'cancelled'
timestamps		

bank_holidays — Country-specific public holidays + org overrides.

Column	Type	Notes
id	uuid	PK

country_code	text	ISO code
organisation_id	uuid	Null = system-wide, non-null = org override
date	date	
name	text	e.g. "Christmas Day"
is_excluded	boolean	Org removed this date from their list

RLS Notes

- Employees: read their own bookings, requests and entitlements only. Create requests (subject to balance/overlap checks in server action). Cannot approve.
- Admins: read bookings for their team (or all, per can_view_all_teams). Approve/reject if can_approve_holidays.
- Owners: full read/write across org.
- All write operations through server actions with permission checks.

Business Rules

To be documented as logic is implemented.

Balance Calculation

```
None
effective_entitlement = pro_rata_amount + adjustment +
carried_over
used = sum(days_deducted) WHERE status IN ('approved', 'pending')
remaining = effective_entitlement - used
```

Note: Pending requests count against balance to prevent overbooking.

Pro-rating (mid-year starters)

```
None
months_remaining = months from start_date to year_end
pro_rata = base_amount * (months_remaining / 12)
```

Rounding: round down to nearest 0.5.

Overlap Detection

- Same employee, overlapping dates, status not 'cancelled' or 'rejected' → block.
- Same team, overlapping dates → warn (informational, not blocking).

Day Counting (days mode)

- Only count days in the employee's working pattern.
- Half days count as 0.5.
- Bank holidays within the range: skip if bank_holiday_handling = 'additional', count if 'deducted'.

Accrual Calculation

```
None
monthly_accrual = base_amount / 12
accrued_to_date = monthly_accrual * months_elapsed
```

Accrual date: 1st of each month, or pro-rated for mid-month starters.

UI Notes

To be documented as screens are built.

Planned Screens

- **Employee holiday dashboard** — balance summary, upcoming bookings, outstanding requests, "Request holiday" button
- **Request/book form** — date picker, absence reason selector, half-day toggles, balance indicator

- **Approval queue** — list of pending requests for approver, approve/reject actions
- **Team calendar** — visual calendar showing who's off (colour-coded by leave type)
- **Admin: holiday settings** — org-wide config (holiday profiles, bank holidays)
- **Admin: entitlements** — table of employees with their entitlements, adjustments, balances
- **Mobile: holiday tab** — balance, upcoming, request